



Post Office Box 2000, Decatur, Alabama 35609-2000

February 2, 2026

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 3
Renewed Facility Operating License No. DPR-68
NRC Docket No. 50-296

Subject: **Licensee Event Report 50-296/2025-002-01**

Reference: **Letter from TVA to NRC, "Licensee Event Report 50-296/2025-002-00," dated October 20, 2025 (ML25293A446)**

The enclosed Licensee Event Report provides details of the inoperability of the BFN Unit 3 Reactor Core Isolation Cooling system for a period exceeding the system's Technical Specification (TS) allowed outage time. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's TS. Revision 1 to this report contains a correction to the Corrective Actions for this condition.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Nuclear Site Compliance Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in blue ink, appearing to read "D. Komm", is written over a horizontal line.

Daniel A. Komm
Site Vice President

Enclosure: Licensee Event Report 50-296/2025-002-01 – Reactor Core Isolation Cooling Inoperable due to a Failed Ramp Generating Signal Converter

U.S. Nuclear Regulatory Commission
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cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Unit 3**

Licensee Event Report 50-296/2025-002-01

**Reactor Core Isolation Cooling Inoperable due to a Failed Ramp Generating Signal Converter
See Enclosed**

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027					
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: pira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.					
1. Facility Name Browns Ferry Nuclear Plant, Unit 3				☒ 050 ☐ 052		52. Docket Number 00296		3. Page 1 OF 6			
4. Title Reactor Core Isolation Cooling Inoperable due to a Failed Ramp Generating Signal Converter											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number	
08	19	2025	2025	- 002 -	01	02	02	2026	N/A	05000XXX	
									Facility Name	Docket Number	
									N/A	05000XXX	
9. Operating Mode 1						10. Power Level 100					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.21(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify here, in abstract, or NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Baruch Calkin, Compliance Engineer								Phone Number (Include area code) 256-278-1031			
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS		
A	BN	SC, CNV	W90	Y	N/A	N/A	N/A	N/A	N/A		
4. Supplemental Report Expected)					15. Expected Submission Date			Month	Day	Year	
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date						N/A	N/A	N/A	
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)											
On August 19, 2025, at 1128 Central Daylight Time (CDT), during the performance of 3-SR-3.5.3.3, RCIC System Rated Flow at Normal Operating Pressure, the Reactor Core Isolation Cooling system (RCIC) failed to achieve required flow rate and pump discharge pressure due to an issue with the governor control system. Upon investigation, it was determined that the Unit 3 RCIC Ramp Generating Signal Converter (RGSC) module had failed on July 20, 2025. On August 23, 2025, at approximately 0108 CDT, the Unit 3 RCIC was declared operable following replacement of the RGSC and successful Cold Quick Start Post Maintenance Testing. The apparent cause of this condition was accumulated stress over time on the operational amplifiers leading to failure of the RGSC. The corrective actions for this condition are to replace the Unit 3 RCIC RGSC, to replace aging RGSCs and Electro-Governor Magnetic Pickups at the earliest opportunity, and to create new PMs to periodically replace similar components on the Unit 1, 2, and 3 RCIC and High Pressure Coolant Injection systems.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 3	☒ 050	2. DOCKET NUMBER 00296	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 002	REV NO. - 01

NARRATIVE**I. Plant Operating Conditions before the Event**

At the time of discovery, Browns Ferry Nuclear (BFN) Unit 3 was in Mode 1 at approximately 100 percent power.

II. Description of Event**A. Event Summary**

On August 19, 2025, at 1128 Central Daylight Time (CDT), during the performance of 3-SR-3.5.3.3, RCIC System Rated Flow at Normal Operating Pressure, the Reactor Core Isolation Cooling system (RCIC) [BN] failed to achieve required flow rate and pump discharge pressure due to an issue with the governor control system. Upon investigation, it was determined that the Unit 3 RCIC Ramp Generating Signal Converter (RGSC) [SC][CNV] module had failed on July 20, 2025. On August 23, 2025, at 0108 CDT, the Unit 3 RCIC was declared operable following replacement of the RGSC and successful Cold Quick Start Post Maintenance Testing (PMT).

The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(i)(B), as a condition that was prohibited by the plant's Technical Specifications (TS).

B. Status of structures, components, or systems (SSCs) that were inoperable at the start of the event and that contributed to the event

There were no SSCs whose inoperability contributed to this event.

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NARRATIVE**C. Dates and approximate times of occurrences**

<u>DATE AND APPROXIMATE TIMES (CDT)</u>	<u>OCCURRENCE</u>
July 20, 2025 1154	Voltage drops on Unit 3 RCIC RGSC, resulting in a deficient condition.
August 19, 2025 1128	During performance of 3-SR-3.5.3.3 RCIC failed to achieve required flow rate and pump discharge pressure due to an issue with the governor control system.
August 23, 2025 0108	Declared Unit 3 RCIC operable following replacement of RGSC and successful Cold Quick Start PMT.

D. Manufacturer and model number of each component that failed during the event

The RGSC that failed was manufactured by the Woodward Governor Company, part number 9903-015.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

This condition was discovered when RCIC failed to achieve the required flow rate and pump discharge pressure during the performance of 3-SR-3.5.3.3, RCIC System Rated Flow at Normal Operating Pressure.

G. The failure mode, mechanism, and effect of each failed component

The RGSC was an obsolete component under a run-to-failure maintenance strategy, which most likely failed due to stress over time on the operational amplifiers.

H. Operator actions

There were no operator actions associated with this event.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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NARRATIVE

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

III. Cause of the event

The direct cause of this condition was a failure of the Unit 3 RCIC RGSC.

A. Cause of each component or system failure or personnel error

The apparent cause of the Unit 3 RCIC RGSC failure was accumulated stress over time on the operational amplifiers [AMP].

Additionally, the failure to implement Preventive Maintenance (PM) to replace the RGSC on a frequency conducive to preventing age-related failures was identified as a contributing cause.

B. Cause(s) and circumstances for each human performance related root cause

The RCIC System Engineer failed to follow the system monitoring plan for RCIC.

IV. Analysis of the event

The RCIC provides makeup water to the reactor vessel during shutdown and isolation from the main heat sink to supplement or replace the normal makeup sources and operates automatically in time to obviate any requirement for the Core Standby Cooling Systems.

Unit 3 TS Limiting Condition of Operability (LCO) 3.5.3 requires, in Mode 1 or in Modes 2 and 3 when reactor steam dome pressure is greater than 150 psig, that RCIC be operable. TS LCO 3.5.3 Conditions A.1 and A.2 require, when RCIC is inoperable, that the High Pressure Coolant Injection system (HPCI) [BJ] be immediately verified operable by administrative means; and that the RCIC system be restored to operable status within 14 days or in accordance with the Risk Informed Completion Time program. A Past Operability Evaluation determined that this condition existed since July 20, 2025, excluding the period from August 5, 2025, at 0132 CDT to August 9, 2025, at 0801 CDT when the Unit 3 reactor steam dome pressure was less than 150 psig due to an ongoing Unit 3 maintenance outage, which exceeded the TS LCO 3.5.3 A.1 and A.2 required completion times.

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NARRATIVE

Finally, TS LCO 3.0.4 states that when an LCO is not met, entry into a Mode or other specified condition in the Applicability shall only be made when the associated actions to be entered permit continued operation in the Mode or other specified condition in the Applicability for an unlimited period of time. On August 9, 2025, BFN, Unit 3 entered a TS 3.5.3 Applicable Mode when TS 3.5.3 Required Actions were not met.

V. Assessment of Safety Consequences

A Probabilistic Risk Analysis determined that this condition was of very low safety significance. The health and safety of the public was not affected.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The periods where the Unit 3 RCIC was inoperable included several windows where the Unit 3 HPCI was concurrently inoperable.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shut down.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

The RCIC system is not credited in the BFN Final Safety Analysis Report. This event did not affect the operability of any safety system.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's Corrective Action Program (CAP) under Condition Report 2033486.

**LICENSEE EVENT REPORT (LER)
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NARRATIVE**A. Immediate Corrective Actions**

The Unit 3 RCIC RGSC was replaced under Work Order 125559901.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

The corrective actions for this event are as follows:

1. To create new PMs to periodically replace the HPCI and RCIC RGSCs and Electro-Governor Magnetic Pickup (EG-M).
2. To replace the following components at the earliest opportunity due to age:
 - Unit 1 HPCI EG-M
 - Unit 1 HPCI RGSC
 - Unit 1 RCIC EG-M
 - Unit 2 RCIC EG-M
 - Unit 2 RCIC RGSC

VII. Previous Similar Events at the Same Site

A review of the BFN CAP and Licensee Event Reports for Units 1, 2, and 3 revealed no other instances within the past three years of RCIC inoperability due to RGSC failure.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.